

Graphs Recap

Average to High Minimum Degree Every non-empty graph G has a subgraph $H \subset G$ such that $\delta(H) \geq \frac{1}{2}\bar{d}(G)$. Repeatedly remove vertices with degree less than the average degree, in total we remove less than the total number of edges.

Linear Programming

Linear Program Maximise/Minimise $c^T x$, subject to linear constraints such as $Ax \leq b, Ax = b, x \geq 0$ where x is a vector.

Feasible Solution A vector x which satisfies all constraints. The set of feasible solutions is the feasible region.

Optimal Solution If we want to maximise $c^T x$, then x^* is optimal if $c^T x^* \geq c^T x$ for all feasible x . If the feasible region is non-empty, either an optimal solution exists or the objective is unbounded above.

Visualising Feasible Region For a 2D problem, the feasible region is a convex polygon.

Convexity A set $C \subseteq \mathbb{R}^n$ is convex if $\forall x, y, \in C, \lambda \in [0, 1], \lambda x + (1 - \lambda)y \in C$. The line connecting x and y stays strictly within C . The intersection of convex sets is convex.

Half-Space $a_1 x_1 + \dots + a_n x_n \leq b$ defines a half-space $H := \{x \in \mathbb{R}^n : a^T x \leq b\}$ and is convex. The feasible region is an intersection of half-spaces corresponding to each constraint.

Objective Hyperplanes The objective $c^T x$ gives a family of parallel hyperplanes $c^T x = \alpha$.

Extreme Point $x \in P$ where it cannot be written as $\lambda y + (1 - \lambda)z$ (does not lie on a line segment). An optimal solution to the maximisation problem can be found at an extreme point of P .

Unboundedness and Infeasibility An LP may be infeasible: no feasible x exists; unbounded: objective tends to infinity; or have at least one optimal solution.

LP Forms Inequality Form: $Ax \leq b, x \geq 0$; Standard (Equality) Form:

$Ax = b, x \geq 0$.

Slack Variables Convert inequalities to equalities; for $Ax \leq b, x \geq 0$, convert to $Ax + s = b, x \geq 0, s \geq 0$. Augment A' with I , and x' with s .

Squeeze To convert equality to inequality, simply squeeze using two inequalities. $Ax = b$ becomes $Ax \leq b, -Ax \leq -b$.

Farkas Lemma Applies to both equality and inequality versions; Exactly one of the following holds:

- $\exists x \geq 0, Ax = b$ or $Ax \leq b$
- $\exists y, y^T A \geq 0, y^T b < 0$

The second statement is a certificate for infeasibility.

Cones $\text{cone}(a_1, \dots, a_n) = \{\sum_{j=1}^n x_j a_j : x_j \geq 0\}$. This contains all nonnegative linear combinations which are closed under scaling by a positive factor. Feasible points lie within this cone. If $b \notin C$, then we have $\exists y, y^T x \geq 0, \forall x \in C, y^T b < 0$. The half cone and b lie on opposite sides of the line $y^T x = 0$, and the system is infeasible.

Optimality Criterion For a bounded LP, an optimal solution occurs at an extreme point of the feasible region. Then x^* optimal $\Rightarrow c^T x^* \geq c^T x$ for all corners x .

Duality

Orthogonality For $u, v \in \mathbb{R}^n, u^T v = 0 \Rightarrow u \perp v$.

Annihilators For a subspace $U \subseteq \mathbb{R}^n$, its annihilator is

$U^\perp = \{y \in \mathbb{R}^n : y^T u = 0, \forall u \in U\}$.

Primal LP $\max c^T x, Ax \leq b, x \geq 0$.

Dual LP $\min b^T y, A^T y \geq c, y \geq 0$. Derived using $c^T x \leq y^T Ax \leq y^T b$.

Primal to Dual LP Flip objective and constraints, then transpose weight matrix A .

Weak Duality If x is primal-feasible and y is dual-feasible, $c^T x \leq b^T y$, $\max(\text{primal}) \leq \min(\text{dual})$.

Strong Duality Let P^*, D^* be the optimal primal and dual values if they exist. Weak duality says $P^* \leq D^*$ while strong duality says $P^* = D^*$.

Integer Programming Additional constraint that entries in x are integers.

LP Relaxation For IP, solve the relaxed problem for nonnegative real numbers. This upper bounds the optimal integer value for max.

Convex Programming Replaces linearity with convexity. LP is a special CP problem.

Independence, Matchings and Covers

Matching Subset of edges where no two edges share a common endpoint. A perfect matching covers all vertices. A maximal matching cannot be enlarged by adding an edge. A maximum matching has maximum possible size. Not all maximal matchings are maximum. The matching number $\nu(G)$ is the maximum size of a matching in G .

Vertex Cover Set of vertices that contains at least one endpoint of every edge. The vertex cover number $\tau(G)$ is the minimum size of a vertex cover of G .

Proposition $\tau(G) \geq \nu(G)$. For bipartite graphs, equality holds.

Clique Subset of vertices that induces a complete subgraph. The clique number $\omega(G)$ is the size of the largest clique in G .

Independent Set Set of vertices S where no two vertices in S are adjacent. The independence number $\alpha(G)$ is the maximum size of an independent set in G .

Cliques and Independence $\alpha(G) = \omega(\overline{G}), \omega(G) = \alpha(\overline{G})$.

Kneser Graph $KG(n, k)$ is the graph where vertices are k -subsets of $[n]$ and two vertices are adjacent if their sets are disjoint. $\alpha(KG(n, k)) = \binom{n-1}{k-1}$.

Independence and Vertex Cover For n vertices, $\alpha(G) + \tau(G) = n$.

Line Graph Vertices are edges of G , while two vertices of $L(G)$ are adjacent if the corresponding edges in G share an endpoint. Then $\nu(G) = \alpha(L(G))$.

Maximum Matching LP Maximise $\sum_{e \in E} x_e$, where $\sum_{v \in e} x_e \leq 1, x_e \in \{0, 1\}$. This selects disjoint edges, using x_e for each edge.

Minimum Vertex Cover LP Minimise $\sum_{v \in V} y_v$, where $y_u + y_v \geq 1, y_v \in \{0, 1\}$. This ensures each edge is covered, using y_v for each vertex.

Dual LPs Let B be the vertex-edge incidence matrix. Then the primal LP is the fractional matching problem while the dual LP is the fractional vertex cover problem. Weak duality implies $\sum x_e \leq \sum y_v$, which gives us $\nu(G) \leq \tau(G)$.

Bipartite Matching

Bipartite Graphs Graph where $V = A \sqcup B$ (disjoint union). Equivalently, G has no odd cycle or $\chi(G) \leq 2$.

Perfect Matching A matching is perfect to A if every vertex of A is matched.

$|A| \leq |B|$ is necessary.

Hall's Marriage Theorem A bipartite graph contains a matching perfect to A iff the marriage condition holds; $\forall S \subseteq A, |N(S)| \geq |S|$. Fails on infinite graphs.

Regular Bipartite Graphs Every d -regular bipartite graph has a perfect matching using Hall's theorem with $d|X| \leq d|N(x)|$.

Edge Decomposition The edges of a d -regular bipartite graph can be decomposed into d perfect matchings; repeatedly remove a perfect matching to get a $d - 1$ -regular graph.

Triesch's Conjecture If $\forall S \subseteq A, |N_G(S)| \geq |N_H(S)|$, then $m(G) \geq m(H)$ where m is the number of perfect matchings.

Hall's using Farkas' Introduce $x_e \geq 0$ for each edge and consider $\sum x_e = 1, \forall a \in A, \sum x_e \leq 1 \forall b \in B$. Infeasibility occurs when for real y_a and nonnegative $z_b, y_a + z_b \geq 0 \forall ab \in E$ and $\sum y_a + \sum z_b < 0$, giving $|N(S)| < |S|$.

König's Theorem For any bipartite graph, $\nu(G) = \tau(G)$.

General Matching

Alternating Path (wrt M) Path such that v_1 is unmatched, and the edges alternate between not in M and in M . An augmenting path also ends at an unmatched vertex. Inverting an augmenting path increases the matching size.

Berge's Theorem A matching M is maximum iff there exists no augmenting path with respect to M .

Odd Component There are $o(H)$ components with an odd number of vertices.

Tutte's Theorem A graph has a perfect matching iff $\forall U \subseteq V, o(G[V \setminus U]) \leq |U|$.

Tutte-Berge Formula $\nu(G) = \frac{1}{2} \min_{U \subseteq V} (|U| - o(G[V \setminus U]) + n)$; alternatively, $2\nu(G) = n - \max_{U \subseteq V} (o(G[V \setminus U]) - |U|)$.

Deficiency $(U) = o(G[V \setminus U]) - |U|$. If positive, then at least (U) vertices must remain unmatched since we cannot fully match odd components.

Matching ILP Introduce a binary variable x_e for each edge. Then the constraints are $\sum_{v \in e} x_e \leq 1$ and we want to maximise $\sum_{e \in E} x_e$.

Berge's and LP An augmenting path is a feasible direction increasing the objective value. Berge's states local optimality (no augmenting path) implies global optimality.

Tutte's and LP Tutte's inequalities characterise when a solution exists for a perfect matching.

Tutte-Berge as Duality The primal problem is the maximum matching size, while the dual problem is maximising the deficiency. Tutte-Berge express the strong duality of these problems.

Matching Polytope $P(G) = \text{conv}\{\chi(M) : M \text{ is a matching of } G\} \subseteq \mathbb{R}^E$, where $\chi(M)$ is the incidence vector of M (for edges) and conv is the convex hull. Each vertex is a matching and linear optimisation over it is the maximum matching problem. For a bipartite graph, all vertices are integral.

Big Picture (Combinatorics v.s LP)

- Matching $\Leftrightarrow$ Integer Feasible Solutions
- Augmenting Path $\Leftrightarrow$ Improving Direction
- Berge's Theorem $\Leftrightarrow$ Optimality Condition
- Tutte's Condition $\Leftrightarrow$ Feasibility Inequalities
- Tutte-Berge $\Leftrightarrow$ Strong Duality
- Matching Polytope $\Leftrightarrow$ Convex Hull of Matchings

Connectivity

Separating Set / Vertex Cut A set $S \subset V(G)$ such that $G - S$ has more than one connected component.

Cut Vertex Separating set consisting of a single vertex.

Vertex Connectivity $\kappa(G)$, minimum size of $S \subset V(G)$ such that $G - S$ is disconnected or has only one vertex.

k-Connected $\kappa(G) \geq k$. Removing $k - 1$ vertices leaves the graph connected.

2-Connected Graphs Every 2-connected graph has a cycle.

Connectivity and Minimum Degree $\kappa(G) \geq \delta(G)$.

Mader's Theorem Every graph of average degree at least $4k$ contains a k -connected subgraph.

Edge Connectivity A graph is ℓ -edge-connected if for every subset of edges with $|F| < \ell, G - F$ remains connected. $\kappa'(G)$ is the maximum such ℓ .

Edge Cuts and Bridges Every minimal disconnecting set is an edge cut. Any edge whose removal disconnects G is a cut edge or bridge. $\kappa'(G) = 1$ iff G has a bridge.

Cut Relations $\kappa(G) \leq \kappa'(G) \leq \delta(G)$.

Chartrand-Harary Theorem For all $0 < a \leq b \leq c$, there exists G such that $\kappa(G) = a, \kappa'(G) = b, \delta(G) = c$.

Whitney's Theorem A graph is 2-connected iff for every pair of vertices there exists two internally-disjoint $u - v$ paths or they belong to a cycle.

A-B Path Path such that $V(P) \cap A = \{x_0\}, V(P) \cap B = \{x_k\}$.

Separating Sets A set separates A and B if every $A - B$ path contains a vertex or edge from X .

Cut Vertex Vertex separating two other vertices in the same component.

Cut Edge Edge separating its two endpoints.

Menger's Theorem The minimum number of vertices separating A from B is equal to the maximum number of vertex-disjoint A - B paths.

Stronger Lemma Let k be the minimum number of vertices separating A and B . If $\mathcal{P}$ is a set of fewer than k vertex-disjoint A - B paths, then there exists a larger set $\mathcal{Q}$ of vertex-disjoint A - B paths whose set of end vertices properly contains the set of ends of $\mathcal{P}$.

Corollaries of Menger's Theorem If a and b are not adjacent, then the minimum number of vertices separating them equals the maximum number of independent (internally vertex-disjoint) a - b paths. The minimum number of edges separating a and b equals the maximum number of edge-disjoint a - b paths.

Global Menger's Theorem A graph is k -vertex-connected iff every pair of vertices is joined by k internally-disjoint paths. A graph is k -edge-connected iff every pair of vertices is joined by k edge-disjoint paths.

Packing Problem Find as many internally vertex-disjoint a - b paths as possible. Let $\mathcal{P}$ be the set of all a - b paths. Introduce x_p which is binary for each path.

Packing Primal LP Maximise $\sum_{p \in \mathcal{P}} x_p$, subject to $\sum_{v \in P} x_p \leq 1$ for all $v \in V(G) \setminus \{a, b\}$, $x_p \geq 0$. This asks for a fractional packing of paths.

Covering Dual LP Introduce $y_v \geq 0$ for each $v \in V(G) \setminus \{a, b\}$. Minimise $\sum_{v \in V(G) \setminus \{a, b\}} y_v$ subject to $\sum_{v \in V(P)} y_v \geq 1$ for all paths. y_v is the weight assigned to vertex v , every a - b path must be hit by a non-zero weight.

Separating Sets If $X \subset V(G) \setminus \{a, b\}$ separates a and b , define $y_v = 1$ if $v \in X$. Then every a - b path meets X so the dual constraints are satisfied. Integral dual solutions correspond to separating vertex sets, so the dual objective is the size the separator. The dual LP is a covering problem.

Menger's as Duality If both LPs are feasible, then Menger's is a strong duality theorem.

Flows

Flow Network Directed graph with a source (all out-edges) and sink (all in-edges), as well as a non-negative capacity on each edge.

Flow A map $f : E(D) \rightarrow \mathbb{R}_{\geq 0}$ satisfying capacity constraints ($0 \leq f_{u \rightarrow v} \leq c_{u \rightarrow v}$) for every edge, as well as flow conservation ($\sum_{u: u \rightarrow v} f_{u \rightarrow v} = \sum_{w: w \rightarrow v} f_{w \rightarrow v}$) for every non source or sink vertex. **Size of Flow** $|f| = \sum_{v: s \rightarrow v} f_{s \rightarrow v} = \sum_{v: v \rightarrow t} f_{v \rightarrow t}$.

Max-Flow LP Find a flow with maximum possible size. Maximise $\sum_{(s,v) \in E} f_{sv} - \sum_{(v,s) \in E} f_{vs}$ (value of flow), under the constraints that $\sum_{(u,v) \in E} f_{uv} - \sum_{(v,w) \in E} f_{vw} \leq 0, \forall v \in V \setminus \{s, t\}$, $-\sum_{(u,v) \in E} f_{uv} + \sum_{(v,w) \in E} f_{vw} \leq 0, \forall v \in V \setminus \{s, t\}$, $0 \leq f_{uv} \leq c_{uv}, \forall uv \in E$ (flow conservation).

Cut Partition such that s and t fall into different components.

Capacity of Cut $c(S, T) = \sum_{u \in S, v \in T, u \rightarrow v \in E(D)} c_{u \rightarrow v}$, only forward edges.

Flow Across Cut The net flow across a cut is $\sum_{u \in S, v \in T, u \rightarrow v \in E(D)} f_{u \rightarrow v} - \sum_{u \in S, v \in T, v \rightarrow u \in E(D)} f_{v \rightarrow u}$; the net flow going from nodes in S to nodes in T . The net flow across any cut is equal to the global flow. For every flow and cut, $|f| \leq c(S, T)$.

Min-Cut Cut minimising $c(S, T)$. By $|f| \leq c(S, T)$,

$\max_f |f| \leq \min_{(S, T)} c(S, T)$.

Min-Cut LP Introduce real z_v for each vertex. Then minimise

$\sum c_{u \rightarrow v} (z_v - z_u)$, subject to $z_s \leq 0, z_t \geq 1, z_v - z_u \geq 0$. z_v is the "side" of the cut containing v , with vertices with smaller z_v lying on the s -side.

Max-Flow Min-Cut Theorem The Max-Flow and Min-Cut LPs form a Primal-Dual Pair. $\max_f |f| = \min_{(S, T)} c(S, T)$.

Integral Flow Theorem If all capacities are integers, there exists a max-flow such that all edge flows are also integers.

Totally Unimodular Matrix Every square submatrix has determinant in $\{0, -1, 1\}$. They imply integrality of LP vertices when the RHS is integral.

Integer Vertices If A is TU and b is an integer vector, the polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ has only integer vertices. Then every bounded LP of the form $\max\{c^T x : Ax \leq b\}$ has an optimal solution at an integral vertex.

Incidence Matrix The $|V| \times |E|$ matrix where an entry $A_{v,e}$ is 1 if v is an endpoint of e . Each column has exactly two 1s.

Bipartite Incidence Matrices Every bipartite graph has a TU incidence matrix.

Signed Incidence Matrix For directed graph, source has -1 , sink has 1. It is TU for every directed graph.

Appending Preserves TU If A is TU, the block matrix $(A, I, -I)^T$ is also TU.

Max-Flow LP in Matrix Form Let A be the signed incidence matrix of the flow network. Let A_0 be A with the rows for s and t removed. Flow conservation is $A_0 x = 0$ and capacity constraints are $0 \leq x \leq c$. The objective is to maximise $\max\{w^T x : A_0 x = 0, 0 \leq x \leq c\}$ where $w^T x$ measures the flow value. If c is integral, then the RHS vector is integral, giving the integral flow theorem.

Packing Problem Given ground set E and family $\mathcal{F}$ of feasible subsets of E , find a maximum collection of pairwise disjoint members of $\mathcal{F}$.

Covering Problem Find a minimum set of elements of E that intersects every member of $\mathcal{F}$.

Approximating Packing-Covering

- Relax integrality by allowing fractional solutions.
- Use LP strong duality.
- The gap between the integer optimum and LP optimum is the integrality gap.
- The gap vanishes when the constraint matrix is TU and all capacities are integers.

b-Fold Packing Relaxed packing problem where we allow each element of E to belong to at most b chosen members of $\mathcal{F}$.

Max-Flow Min-Cut Tool

- Model the packing problem as a max-flow problem.
- Model the covering problem as a min-cut problem.
- Max-Flow Min-Cut gives equality of optima.
- Integral Flow Theorem (from TU) ensures an integer optimal flow exists.

Reduction to Network Flow

- Model combinatorial problem as a flow network.
- identify max flow with one combinatorial quantity.
- Identify min cut with the dual quantity.
- Apply Max-Flow Min-Cut (and integrality when needed).

Menger's (Directed, Edge-Disjoint)

- Consider a flow network where every edge has capacity 1.
- By Integral Flow Theorem, the max flow is an integer that can be decomposed into unit flows along edge-disjoint x - y paths.
- A min-cut then has capacity equal to the number of edges that must be removed to disconnect the graph.
- Then by duality we have this equal to the maximum number of edge-disjoint paths.

Menger's (Directed, Vertex-Disjoint)

- Split vertices into v^-, v^+ . Then $v^- \rightarrow v^+$ has capacity 1, while $u^+ \rightarrow v^-, v^+ \rightarrow w^-$ has capacity ∞ .
- A min-cut corresponds to a vertex separator, so we also get this to be equal to the number of vertex-disjoint paths.

Menger's (Undirected) Replace each undirected edge with two directed edges, then apply the directed version.

König's Theorem

- Construct the flow network where the source is connected to A and the sink to B with capacity 1. Then have edges from A to B with capacity ∞ .
- Max-Flow = $\nu(G)$: An integral flow sends 0 or 1 on each source or sink edge. The edges across the middle carrying flow 1 form a matching. Any matching gives a feasible integral flow.

- Min-Cut = $\tau(G)$: A finite cut cannot use any ∞ -edge. So the cut edges are a subset of the source or sink edges. This cuts all s - t paths iff $A' \cup B'$ is a vertex cover. The capacity is $|A'| + |B'|$.
- Max-Flow Min-Cut gives equality.

Hypergraphs

Hypergraph Consists of vertex set and edge set where each edge is a subset of the vertex set. An ordinary graph has edge size 2.

Uniform Hypergraph A hypergraph is k -uniform if every edge has k vertices.

Complete k-uniform Hypergraph $K_n^{(k)}$, with vertex set $[n]$, edge set $E = \binom{[n]}{k}$.

Hypergraph Matching Collection of pairwise disjoint edges. It is perfect if every vertex belongs to exactly one edge in the matching. For a k -uniform hypergraph, a perfect matching has exactly n/k edges and exists only if $k|n$.

Baranyai's Theorem If $k|n$, then the complete k -uniform hypergraph $K_n^{(k)}$ can be decomposed into perfect matchings.

Stronger Claim For any $1 \leq l \leq n$, there exists a collection $\mathcal{A}_1, \dots, \mathcal{A}_m$ of m -partitions of $[l]$ (m disjoint subsets) such that each subset of $[l]$ appears exactly $\binom{n-l}{k-|S|}$ times across all partitions.

Flow Numbers

Nowhere-Zero k-Flows Map $f : E(G) \rightarrow \{1, 2, \dots, k-1\}$ satisfying flow conservation. The graph must be bridgeless.

Flow Number $\varphi(G)$, smallest k for which a nowhere-zero k -flow exists.

2-Flow G has a 2-flow iff every vertex must have an even degree.

3-Flow G has a 3-flow iff it is bipartite.

Complete Graph For all even $n > 4$, $\varphi(K_n) = 3$.

4-Flow Every 4-edge-connected graph has a 4-flow.

Theorem (Seymour's) Every bridgeless graph has a 6-flow. $\varphi(G)$ is finite iff G is bridgeless.

Tutte's Conjectures

- 5-Flow: Every bridgeless multigraph has a 5-flow.
- 4-Flow: Every bridgeless multigraph with no Petersen-graph minor has a 4-flow.
- 3-Flow: Every multigraph with no cut of size 1 or 3 has a 3-flow.

Flows over Abelian Groups A nowhere-zero M -flow is a map $f : E(G) \rightarrow M \setminus \{0\}$ satisfying flow conservation. An example is $M = \mathbb{Z}_k = \mathbb{Z} \setminus k\mathbb{Z}$ (modulo k).

Theorem (Tutte) Nowhere-zero k -flow exists iff nowhere zero $\mathbb{Z}_k$ -flow exists.

Notation

- $\delta(G)$: Minimum Degree
- $\Delta(G)$: Maximum Degree
- $\nu(G)$: Matching number / Maximum matching size
- $\tau(G)$: Vertex cover number / Minimum vertex cover
- $\omega(G)$: Clique number / Maximum clique size
- $\alpha(G)$: Independence number / Maximum independent set size
- $o(H)$: Number of odd components
- $\kappa(G)$: Vertex connectivity / Minimum size of vertex cut
- $\kappa'(G)$: Edge connectivity / Minimum size of edge cut
- $\varphi(G)$: Flow number

Tutorial Stuff

T1Q2 If G is regular, then so is $L(G)$. G is connected iff $L(G)$ is connected.

T1Q3 Girth is the length of the shortest cycle in the graph. A k -regular graph with girth 5 has at least $k^2 + 1$ vertices.

T2Q1 $\alpha(G) \geq \frac{|V(G)|}{\Delta(G)+1}$, $\tau(G) \leq |V(G)| \frac{\Delta(G)}{\Delta(G)+1}$.

T3Q2 Every 3-regular graph with no cut edge has a perfect matching.

T4Q1 For $k \geq 2$, a connected k -regular bipartite graph is 2-connected.

T4Q2 If a graph is 3-regular, then $\kappa(G) = \kappa'(G)$.

T4Q4 G is 2-connected iff for any x, y, z , there is a path from x to y containing z .